

* Calibrating distances:

→ Hubble Distances

- "Near": Galaxies are moving away
- "far" → Cosmology / GR

→ Near: $V_{\text{obs}} = V_{\text{Hubble}} + V_{\text{pert.}}$

$$V_{\text{Hubble}} = H_0 D \leftarrow \text{Distance (near by - by)}$$

↑
Hubble's Const.

V_{pert} : local motion

→ Clusters "bound" by gravity: attractive forces

ex: M31 + MW (local group)

* $V_{\text{Hubble}} = H_0 D \approx c z$

z : redshift-parameter.

* Galaxy counts:

→ Similar IMF: $\xi(M) dM$

$$N_{\star} = \int \xi(M) dM$$

$$M_{\text{Vol}} = \int M \xi(M) dM$$

→ "Find" same for galaxies

• Luminosity, Distance etc...

→ $N_{\text{gal}}(< F)$

→ Similar to $\xi(M)$: $\phi(L) \leftarrow$ Galaxy Luminosity function

$$[\phi(L)] = \frac{N_{\text{gal}}}{\text{Vol}}$$

$$N_{\text{gal}} \sim \phi(L) \text{Vol}$$

$$N(> F) = \phi(L) \frac{4\pi}{3} D^3$$

$$F = \frac{L}{4\pi D^2}$$



$$D = \sqrt{\frac{L}{4\pi F}}$$

$$N(>F) = \phi(L) \frac{4\pi}{3} \left(\frac{L}{4\pi F} \right)^{3/2}$$

$$\phi? : \quad \xi(M) \sim \left(\frac{M}{M_*} \right)^{-\alpha} \quad * \text{ Empirical}$$

near center $\alpha \approx 1.7$

outer $\alpha \approx 2.35$

$$\phi(L) = \frac{\phi_*}{L_*} \left(\frac{L}{L_*} \right)^\alpha e^{-L/L_*} \quad *$$

α : depends on where we're looking

ϕ_* : Normalization const.

B-Bed:

$$\alpha: -2.07$$

$$\phi_*: 24 \times 10^{10} L_\odot$$

$$L_*: 0.047 \text{ Mpc}^{-2}$$

Cosmic density of universe: $\xi(M) dM$

$\phi(L) dL$: $\frac{\# \text{ of gal.}}{\text{Vol. in}} \left(\text{which have } L_{\text{min. between } L \text{ and } L + dL} \right)$

$$\begin{aligned} N_{\text{gal}} &= \int_a^b \phi(L) dL \\ &= \int_a^b \frac{\phi_*}{L_*} \left(\frac{L}{L_*} \right)^\alpha e^{-L/L_*} dL \end{aligned}$$

$$\begin{aligned} I(R) &= I_c e^{-R/R_c} * \\ &\sim \frac{L}{4\pi D^2} \end{aligned}$$

Notes: $\frac{\phi_*}{L_*} \int_1^\infty \left(\frac{L}{L_*} \right)^\alpha e^{-L/L_*} dL$

$$u = L/L_* \Rightarrow du = \frac{dL}{L_*}$$

$$= \frac{\phi_*}{L_*} \int_1^\infty u^\alpha e^{-u} L_* du$$

$$* = \phi_* \int_{L/L_*}^\infty u^\alpha e^{-u} du \leftarrow L \quad *$$

$$\Gamma(\underline{z}+1) = \underline{z}! = \int_0^\infty t^{\underline{z}} e^{-t} dt$$

$$\Gamma(\underline{z}+2, L) = \int_L^\infty t^{\underline{z}} e^{-t} dt$$

$$\Lambda_{gal}(L, \infty) = \phi_k \Gamma(\alpha+2, L/L_k) \quad *$$

Luminosity density:

Recall IMF: $\int \xi(M) dM = \frac{\# \text{ stars}}{\text{Vol}}$
 $M \int \xi(M) dM = \frac{(\text{total mass})}{\text{Vol}}$

$\phi(L) dL: \frac{\# \text{ gal}}{\text{Vol}}$

$\int \phi(L) dL = \frac{(\text{total lum.})}{\text{Vol}}$

$$\begin{aligned} L_{\text{tot}} &= \int_L \left(\frac{\phi_k}{L_k} \right) \left(\frac{L}{L_k} \right)^\alpha e^{-L/L_k} dL \\ &= \phi_k \int_L \left(\frac{L}{L_k} \right)^\alpha \left(\frac{L}{L_k} \right) dL \leftarrow dL = \frac{dL}{L_k} \\ &= \phi_k \int_L \left(\frac{L}{L_k} \right)^{\alpha+1} e^{-L/L_k} dL \\ &\quad \uparrow \\ &\quad \alpha+1 \leftarrow \text{gives neg} \end{aligned}$$

$$\begin{aligned} L_{\text{tot}} &= L_k \phi_k \Gamma((\alpha+1)+1, L/L_k) \\ &= L_k \phi_k \Gamma(\alpha+2, L/L_k) \end{aligned}$$

$$\alpha+2 \leq 0$$

$$\alpha \leq -2$$

* problem: $L \rightarrow 0 \quad \alpha \leq -2$
 $\rightarrow$ Not perfect.

$$\phi(L) dL = \frac{\phi_k}{L_k} \left(\frac{L}{L_k} \right)^{-\alpha} dL \quad \left| \begin{array}{l} \alpha = 1.4 \\ L_k = 7 \times 10^9 L_0 \end{array} \right| \quad \left| \begin{array}{l} \phi_k = 0.0075 \\ M_{pc}^{-3} \end{array} \right|$$

$$L_1 = 10^9 L_0$$

$$L_2 = 10^{11} L_0$$

$$-1.4 + 1 = -0.4$$

PS6 Q11

$$V_{pec} = 300 \text{ km/s}$$

$$* V_{obs} = V_{Hubble} + \underline{V_{pec}}$$

$$V_{Hs} = \underline{H_0 D} \simeq cz$$

→ Deflect: comes from V_{pec}

$$\left(\frac{V_{pec}}{V_{Hs}} \right) < .1$$

H_0 : const.

$$\left(\frac{300}{H_0 D} \right) < .1$$

$$(1) \left(\frac{300}{H_0 D} \right) < .1$$

$$(2) \frac{300}{cz} < .1$$

$$\boxed{\left(\frac{300}{cz} \right) < .1}$$

$$300 \text{ km/s} = 3 \times 10^5 \text{ m/s}$$

$$c = 3 \times 10^8 \text{ m/s}$$

$$\frac{3 \times 10^5}{3 \times 10^8 \cdot z} = \frac{3 \times 10^6}{3 \times 10^8} = .01$$

$$\boxed{z \geq 0.01}$$

PS5 Q2 $M_V \approx -6$

trial: $V = 22$

Hubble: $V = 26$

$$m - M = 5 \log(D) - 5$$

solve for D

$$\frac{m - M + 5}{5} = \log_{10}(D)$$

$$D = 10^{.2(m - M_V + 5)}$$

$$D_{\text{trial}} = 10^{.2(22 - (-6) + 5)}$$

$$D_{\text{hub}} = 10^{.2(26 - (-6) + 5)} *$$

$$D_{\text{hub}} \gg D_{\text{trial}}$$

* better calibration

→ Distance chapter *

PS5 Q3

$$\frac{L_2}{4\pi d^{10} L_\odot} = \left(\frac{V_{\text{max}}}{200 \text{ km/s}} \right)^4$$

$$m_I = 2.92$$

"Observed" max: $V_m = 226 \text{ km/s}$

$$i = 77^\circ$$

$$M_{I,0} = 4.08$$

$$V_{\text{rec}} = V_{\text{hub}} = H_0 D = 820 \text{ km/s}$$

← V_{obs}



$$\sin(i) = \frac{V_{obs}}{V_{max}}$$

$$V_{max} = \frac{V_{obs}}{\sin(i)}$$

$$\frac{L_2}{L_0} = \left(\frac{(V_{obs} / \sin(i))}{200 \text{ km/s}} \right)^4$$

$$\underbrace{m_I - M_I}_{\substack{\text{distance} \\ \text{modulus}}} = -2.5 \log \left(\frac{L_2}{L_0} \right) \quad \left| \quad I \sim \frac{L}{4\pi D^2} \right.$$

$$= 5 \log(D) - 5$$

D : 17.1 Mpc
compare to

$$V_{rec} = H_0 D'$$

$$D' = \frac{V_{rec}}{H_0} = 11.4 \text{ Mpc}$$

* Renzetti: Idealized M-M: assumed no extinction

$$5 \log(D) - 5 + A_L$$

|||||



i: also

V_{loss} :

3

